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COMMON HALF YEARLY EXAMINATION - 2025

Standard XII

Reg.No. | | | | | | |

BIOLOGY

Part – I (Bio-Botany) 35 marks

Time : 3.00 hrs *Marks : 70*

I. Choose the correct answer: 8 x 1 = 8

1. Male gametes is angiosperms are formed by the division of
   a) microspore mother cell b) microspore
   c) generative cell d) vegetative cell
2. Point mutation involves
   a) Deletion b) Insertion c) Change in single base pair d) duplication
3. In soil water available for plants is
   a) gravitational water b) chemically bound water
   c) capillary water d) hygroscopic water
4. For transformation, micro-particles coated with DNA to be bombarded with gene gun are made up of
   a) silver (or) platinum b) platinum (or) zinc
   c) silicon (or) platinum d) gold (or) tungsten
5. Find out the correctly matched pair.
   a) Rubber – Shorea robusta b) Dye – Indigofera annecta
   c) Timber – Cyperus papyrus d) Pulp – Hevea brasiliensis
6. Consumption of which one of the following foods prevent the kind of blindness associated with Vitamin A deficiency?
   a) Flavr savr b) Canola c) Golden rice d) Bt brinjal
7. Which of the following ecosystem has the highest primary productivity?
   a) Pond ecosystem b) Lake ecosystem
   c) Grassland ecosystem d) Forest ecosystem
8. With respect to Eichhornia
   Statement A : It drains off oxygen from water and is seen growing in standing water.
   Statement B : It is an indigenous species of our country
   a) Statement A is correct and statement B is wrong
   b) Both statements A and B are correct
   c) Statement A is wrong and statement B is correct
   d) Both statements A and B are wrong

II. Answer any four questions 4 x 2 = 8

9. What is Pollen kilt?
10. Write a note on heterosis?
11. Differentiate bio-medicines and botanical medicines?
12. Give four examples of plants cultivated in commercial agroforestry?
13. What is Atavism?
14. Draw the diagram of pollengrain and label the parts?

III. Answer any three questions. (Q.No.19 is compulsory) 3 x 3 = 9

15 Mention the name of man-made cereal. How it is formed?
16. What do you know about the word PBR 332?
17. Write the objectives of Plant Breeding.
18. Write the advantages of Artificial seeds.
19. What are Endemic Centres and Endemic Plants? Write any one example?

IV. Answer all the questions. 2 x 5 = 10

20. a) Enumerate the characteristic features of Entomophilous Flowers *(OR)*
    b) Explain the basic concepts involved in plant tissue culture.
21. a) What is Ten percent law? Explain with example? *(OR)*
    b) What are psychoactive drugs? Add a note Marijuana and Opium?

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Part – II (Bio-Zoology) 35 marks

I. Choose the correct answer: 8 x 1 = 8

1. Down's syndrome is a genetic disorder which is caused by the presence of an extra chromosome number.
   a) 20 b) 21 c) 4 d) 23
2. Assertion (A) : Ovulation is the release of ovum from the Graffian follicle
   Reason (R) : It occurs during the follicular phase of the menstrual cycle.
   a) A and R are true, R is the correct explanation of A
   b) A and R are true, R is not the correct explanation of A
   c) A is true R is false
   d) Both A and R are false
3. An operon is a
   a) Protein that suppresses gene expression
   b) Protein that accelerates gene expression
   c) Cluster of structural genes with related function
   d) Gene that switched other genes on or off
4. Match the pathogens with respective diseases caused by them and select the correct match using the codes given below.

A. Leishmania donavani	–	i)	Amoebiasis
B. Wuchereria bancrofti	–	ii)	Kala–azar
C. Trypanosoma gambiense	–	iii)	Sleeping sickness
B. Entamoeba histolytica	–	iv)	Filariasis

   a) A–ii, B–iv, C–iii D–i b) A–ii, B– iv, c–i, D–iii
   c) A–iii B–i, c–ii, D–iv d) A–i B–iv, C–iii. D–ii
5. The phenomenon of Industrial Melanism demonstrates a) Natural selection b) Induced mutation c) Reproductive isolation d) Geographical isolation
6. Predation and Parasitism are which type of interactions
   a) (+,+) b) (+,0) c) (–,–) d) (+,–)
7. The organisation which published the red list of species is
   a) WWF b) IUCN c) ZSI d) UNEP
8. Average human consumption of Oxygen per day
   a) 1000 L b) 1500 L c) 550 L d) 500 L

II. Answer any four questions. 4 x 2 = 8

9. What is plasmotomy?
10. What is chorionic villus sampling?
11. Why are sex linked recessive characters are more common in the male human beings?
12. Which is called as Brewer's yeast? What are its uses?
13. What are transgenic animals? Give examples
14. Name, the active chemical found in the medicinal plant Rauwolfia Vomitoria. What type of diversity it belongs to?

III. Answer any three questions. (Q.No.19 is compulsory) 3 x 3 = 9

15. Why tRNA is called an adapter molecule?
16. How does Neanderthal man differ from the modern man in appearence?
17. Draw a neat labelled diagram of HIV–virus.
18. Write any three functions of soil.
19. What is plastic waste? What are the remedies for plastic waste.

IV. Answer all the questions. 2 x 5 = 10

20. a) List out the major reproductive events in human beings. *(OR)*
    b) Why do we find a decrease in biodiversity distribution, if we move from the tropics towards the poles.
21. a) List out the salient features of genetic code. *(OR)*
    b) What are the various measures to control Air Pollution.

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